

International Islamic University Chittagong

Department of Science of Hadith and Islamic Studies

Basic Sciences and Engineering

Course Lecture Notes

Course Information

- **Course Code:** BSE-2301
- **Course Title:** Basic Sciences and Engineering
- **Credit Hours:** 3 (Three)
- **Contact Hours:** 3 (Three)
- **Program:** B.A. (Hon's) for all departments of FSIS

Instructor Information

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Academic Year: 2026

Marks Distribution

The total marks for the course are 100.

Assessment Component	Marks
Attendance	10
Assignment / Class Test	10
Mid-Term Examination	30
Final Examination	50
Total	100

Lecture 02

Fractions and Equivalent Fractions

Proper Fractions, Improper Fractions, Mixed Numbers,
Equivalent Fractions and Simplification

1. Common Fractions

A fraction represents a part of a whole, a part of a group, or the result of dividing one number by another.

A fraction is normally written in the form:

$$\frac{a}{b}$$

where:

$$b \neq 0.$$

Definition: Fraction

A fraction is a number that represents one or more equal parts of a whole and is normally written in the form

$$\frac{a}{b},$$

where a is the numerator, b is the denominator, and $b \neq 0$.

For example:

$$\frac{3}{5}$$

means that a whole has been divided into five equal parts and three of those parts are being considered.

2. Numerator and Denominator

Every ordinary fraction contains two important parts.

Consider:

$$\frac{7}{12}$$

Here:

- 7 is the **numerator**.
- 12 is the **denominator**.

Part	Meaning
Numerator	Shows how many parts are being considered.
Denominator	Shows how many equal parts make one whole.

Example

Consider the fraction

$$\frac{5}{8}$$

The denominator 8 tells us that the whole has been divided into eight equal parts.

The numerator 5 tells us that five of those eight parts are being considered.

3. Important Condition of a Fraction

The denominator of a fraction can never be zero.

For example,

$$\frac{5}{0}$$

is undefined.

Important Rule

For any fraction

$$\frac{a}{b},$$

the denominator must satisfy

$$b \neq 0.$$

Division by zero is not defined.

4. Types of Fractions

The three most important forms of fractions for this lecture are:

1. Proper Fractions
2. Improper Fractions
3. Mixed Numbers

5. Proper Fractions

Definition: Proper Fraction

A proper fraction is a fraction in which the numerator is smaller than the denominator.

In general:

$$\frac{a}{b}$$

is a proper fraction when

$$a < b.$$

Examples include:

$$\frac{1}{2}, \quad \frac{3}{5}, \quad \frac{7}{10}, \quad \frac{11}{20}.$$

Every positive proper fraction has a value less than 1.

For example:

$$\frac{3}{4} = 0.75 < 1.$$

Example

Determine whether

$$\frac{7}{9}$$

is a proper fraction.

Here,

$$7 < 9.$$

Therefore, the numerator is smaller than the denominator.

Hence,

$$\frac{7}{9} \text{ is a proper fraction.}$$

6. Improper Fractions

Definition: Improper Fraction

An improper fraction is a fraction in which the numerator is greater than or equal to the denominator.

In general:

$$\frac{a}{b}$$

is improper when

$$a \geq b.$$

Examples include:

$$\frac{7}{4}, \quad \frac{13}{8}, \quad \frac{15}{6}, \quad \frac{9}{9}.$$

An improper fraction normally has a value greater than or equal to 1.

For example:

$$\frac{7}{4} = 1.75 > 1.$$

Also,

$$\frac{9}{9} = 1.$$

Example

Determine whether

$$\frac{17}{12}$$

is proper or improper.

Since

$$17 > 12,$$

the numerator is greater than the denominator.

Therefore,

$$\frac{17}{12} \text{ is an improper fraction.}$$

7. Mixed Numbers

Definition: Mixed Number

A mixed number contains a whole number together with a proper fraction.

Examples include:

$$1\frac{1}{2}, \quad 2\frac{3}{5}, \quad 5\frac{7}{8}.$$

Consider:

$$3\frac{2}{7}.$$

It contains:

$$3$$

whole units and

$$\frac{2}{7}$$

of another whole.

Therefore,

$$3\frac{2}{7} = 3 + \frac{2}{7}.$$

8. Proper, Improper and Mixed Fractions Compared

Type	Condition / Meaning	Examples
Proper Fraction	Numerator is smaller than denominator.	$\frac{3}{5}, \frac{7}{9}$
Improper Fraction	Numerator is greater than or equal to denominator.	$\frac{9}{5}, \frac{13}{8}$
Mixed Number	Contains a whole number and a proper fraction.	$1\frac{4}{5}, 3\frac{2}{7}$

Quick Identification

For positive fractions:

Numerator < Denominator $\Rightarrow$ Proper Fraction

Numerator $\geq$ Denominator $\Rightarrow$ Improper Fraction

A number such as

$$4\frac{2}{5}$$

is a mixed number.

9. Converting Improper Fractions to Mixed Numbers

An improper fraction can be converted into a mixed number by dividing its numerator by its denominator.

Rule: Improper Fraction to Mixed Number

For

$$\frac{a}{b},$$

divide a by b .

$$a = bq + r$$

where:

- q is the quotient,
- r is the remainder.

Then:

$$\frac{a}{b} = q\frac{r}{b}$$

Example 1

Example

Convert

$$\frac{11}{4}$$

into a mixed number.

Divide:

$$11 \div 4.$$

We obtain:

$$11 = 4(2) + 3.$$

Therefore,

$$\frac{11}{4} = 2\frac{3}{4}.$$

Hence,

$$2\frac{3}{4}$$

Example 2

Example

Convert

$$\frac{17}{5}$$

into a mixed number.

Divide 17 by 5:

$$17 = 5(3) + 2.$$

Therefore,

$$\frac{17}{5} = 3\frac{2}{5}.$$

Hence,

$$3\frac{2}{5}$$

Previous-Question-Style Example

Previous-Question-Style Example

Convert

$$\frac{83}{11}$$

into a mixed number.

Divide 83 by 11:

$$83 \div 11.$$

Since

$$11 \times 7 = 77,$$

the remainder is:

$$83 - 77 = 6.$$

Therefore,

$$83 = 11(7) + 6.$$

Hence,

$$\frac{83}{11} = 7\frac{6}{11}.$$

Final Answer:

$$\boxed{7\frac{6}{11}}$$

10. Converting Mixed Numbers to Improper Fractions

A mixed number can be converted into an improper fraction.

Consider:

$$2\frac{3}{5}.$$

The procedure is:

1. Multiply the whole number by the denominator.
2. Add the numerator.
3. Keep the original denominator.

Thus:

$$2 \times 5 = 10,$$

then:

$$10 + 3 = 13.$$

Therefore,

$$2\frac{3}{5} = \frac{13}{5}.$$

Rule: Mixed Number to Improper Fraction

For

$$a\frac{b}{c},$$

use:

$$a\frac{b}{c} = \frac{ac + b}{c}$$

Example 1**Example**

Convert

$$3\frac{2}{7}$$

into an improper fraction.

Multiply:

$$3 \times 7 = 21.$$

Add the numerator:

$$21 + 2 = 23.$$

Keep the denominator 7.

Therefore,

$$3\frac{2}{7} = \frac{23}{7}.$$

Hence,

$$\frac{23}{7}$$

Example 2

Example

Convert

$$5\frac{3}{8}$$

into an improper fraction.

$$5 \times 8 = 40.$$

Then,

$$40 + 3 = 43.$$

Therefore,

$$5\frac{3}{8} = \boxed{\frac{43}{8}}.$$

11. Checking Mixed Number Conversion

A useful way to check the answer is to convert the improper fraction back into a mixed number.

For example:

$$4\frac{3}{5} = \frac{4(5) + 3}{5} = \frac{23}{5}.$$

Now divide:

$$23 \div 5 = 4 \text{ remainder } 3.$$

Therefore,

$$\frac{23}{5} = 4\frac{3}{5}.$$

The conversion is correct.

12. Equivalent Fractions

Fractions may look different but still have exactly the same value.

For example:

$$\frac{1}{2}, \quad \frac{2}{4}, \quad \frac{3}{6}, \quad \frac{4}{8}$$

all represent the same quantity.

Definition: Equivalent Fractions

Equivalent fractions are fractions that have different numerators or denominators but represent the same numerical value.

Thus:

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8}$$

13. Generating Equivalent Fractions by Multiplication

Equivalent fractions can be produced by multiplying both the numerator and denominator by the same nonzero number.

Consider:

$$\frac{2}{3}.$$

Multiply numerator and denominator by 2:

$$\frac{2}{3} = \frac{2 \times 2}{3 \times 2} = \frac{4}{6}.$$

Multiply by 3:

$$\frac{2}{3} = \frac{2 \times 3}{3 \times 3} = \frac{6}{9}.$$

Multiply by 4:

$$\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}.$$

Therefore:

$$\frac{2}{3} = \frac{4}{6} = \frac{6}{9} = \frac{8}{12}$$

Equivalent Fraction Rule

For any nonzero number k ,

$$\frac{a}{b} = \frac{a \times k}{b \times k}$$

provided that

$$b \neq 0 \quad \text{and} \quad k \neq 0.$$

Example

Example

Find three fractions equivalent to

$$\frac{3}{5}.$$

Multiply numerator and denominator by 2:

$$\frac{3}{5} = \frac{3 \times 2}{5 \times 2} = \frac{6}{10}.$$

Multiply by 3:

$$\frac{3}{5} = \frac{9}{15}.$$

Multiply by 4:

$$\frac{3}{5} = \frac{12}{20}.$$

Therefore:

$$\frac{3}{5} = \frac{6}{10} = \frac{9}{15} = \frac{12}{20}$$

14. Generating Equivalent Fractions by Division

Equivalent fractions can also be produced by dividing both numerator and denominator by the same common factor.

Consider:

$$\frac{12}{18}.$$

Divide numerator and denominator by 2:

$$\frac{12}{18} = \frac{6}{9}.$$

Then divide by 3:

$$\frac{6}{9} = \frac{2}{3}.$$

Therefore:

$$\boxed{\frac{12}{18} = \frac{6}{9} = \frac{2}{3}}$$

Important

To preserve the value of a fraction, whatever operation is performed on the numerator must also be performed on the denominator.

For example:

$$\frac{3}{5} = \frac{3 \times 4}{5 \times 4} = \frac{12}{20}.$$

Multiplying only the numerator would change the value.

15. Checking Whether Two Fractions Are Equivalent

One useful method is **cross multiplication**.

Suppose:

$$\frac{a}{b} \quad \text{and} \quad \frac{c}{d}.$$

The fractions are equivalent when:

$$ad = bc.$$

Cross-Multiplication Test

$$\frac{a}{b} = \frac{c}{d} \iff ad = bc$$

provided that

$$b \neq 0 \quad \text{and} \quad d \neq 0.$$

Example 1

Example

Determine whether

$$\frac{3}{4} \quad \text{and} \quad \frac{9}{12}$$

are equivalent.

Cross multiply:

$$3 \times 12 = 36$$

and

$$4 \times 9 = 36.$$

Since:

$$36 = 36,$$

the fractions are equivalent.

Therefore,

$$\frac{3}{4} = \frac{9}{12}$$

Example 2

Example

Determine whether

$$\frac{2}{5} \quad \text{and} \quad \frac{7}{15}$$

are equivalent.

Cross multiply:

$$2 \times 15 = 30$$

and:

$$5 \times 7 = 35.$$

Since:

$$30 \neq 35,$$

the fractions are not equivalent.

Therefore,

$$\frac{2}{5} \neq \frac{7}{15}$$

16. Finding a Missing Numerator

Equivalent fractions can be used to find an unknown numerator.

Consider:

$$\frac{3}{7} = \frac{x}{35}.$$

Since:

$$7 \times 5 = 35,$$

the numerator must also be multiplied by 5:

$$3 \times 5 = 15.$$

Therefore:

$$x = 15.$$

Previous-Question-Style Example

Find x :

$$\frac{3}{7} = \frac{x}{35}.$$

Using cross multiplication:

$$3 \times 35 = 7x.$$

Therefore:

$$105 = 7x.$$

Divide both sides by 7:

$$x = 15.$$

Hence,

$$\boxed{x = 15}$$

17. Finding a Missing Denominator

The same idea can be applied when the denominator is unknown.

Previous-Question-Style Example

Find x :

$$\frac{5}{6} = \frac{45}{x}.$$

Cross multiply:

$$5x = 6 \times 45.$$

Therefore:

$$5x = 270.$$

Divide by 5:

$$x = 54.$$

Hence,

$$\boxed{x = 54}$$

18. Lowest Form of a Fraction

Equivalent fractions may be written using very large numbers.

For example:

$$\frac{18}{24} = \frac{9}{12} = \frac{3}{4}.$$

Among these,

$$\frac{3}{4}$$

is the simplest form.

Definition: Lowest Form

A fraction is in its lowest or simplest form when the numerator and denominator have no common factor greater than 1.

For example:

$$\frac{3}{4}$$

is in lowest form because the only common positive factor of 3 and 4 is 1.

19. Greatest Common Factor

The **Greatest Common Factor (GCF)** is the largest positive number that divides two numbers exactly.

Consider 18 and 24.

Factors of 18 are:

$$1, 2, 3, 6, 9, 18.$$

Factors of 24 are:

$$1, 2, 3, 4, 6, 8, 12, 24.$$

The common factors are:

$$1, 2, 3, 6.$$

Therefore:

$$\boxed{\text{GCF}(18, 24) = 6}.$$

20. Simplifying Fractions Using GCF

The most direct method of simplifying a fraction is to divide the numerator and denominator by their GCF.

Rule: Simplification Using GCF

For

$$\frac{a}{b},$$

if

$$g = \text{GCF}(a, b),$$

then:

$$\boxed{\frac{a}{b} = \frac{a \div g}{b \div g}}$$

Example 1

Example

Simplify:

$$\frac{18}{24}.$$

The GCF of 18 and 24 is:

$$6.$$

Therefore:

$$\frac{18}{24} = \frac{18 \div 6}{24 \div 6} = \frac{3}{4}.$$

Hence,

$$\boxed{\frac{3}{4}}$$

Example 2

Example

Simplify:

$$\frac{42}{56}.$$

The GCF of 42 and 56 is:

$$14.$$

Therefore:

$$\frac{42}{56} = \frac{42 \div 14}{56 \div 14} = \frac{3}{4}.$$

Hence,

$$\boxed{\frac{3}{4}}$$

Example 3

Example

Simplify:

$$\frac{36}{48}.$$

The GCF is:

$$12.$$

Therefore:

$$\frac{36}{48} = \frac{36 \div 12}{48 \div 12} = \frac{3}{4}.$$

Final Answer:

$$\boxed{\frac{3}{4}}$$

21. Simplification by Repeated Cancellation

If the GCF is not immediately obvious, a fraction may be simplified step by step.

For example:

$$\frac{48}{72}.$$

First divide by 2:

$$\frac{48}{72} = \frac{24}{36}.$$

Again divide by 2:

$$\frac{24}{36} = \frac{12}{18}.$$

Divide by 3:

$$\frac{12}{18} = \frac{4}{6}.$$

Finally divide by 2:

$$\frac{4}{6} = \frac{2}{3}.$$

Therefore:

$$\frac{48}{72} = \frac{2}{3}$$

Important

Repeated cancellation and the GCF method produce the same final answer.

Using the GCF is normally faster, but repeated cancellation may be easier for beginners.

22. Simplification Using Prime Factors

Another method is to express the numerator and denominator as products of prime factors.

Consider:

$$\frac{30}{42}.$$

Prime factorization gives:

$$30 = 2 \times 3 \times 5$$

and:

$$42 = 2 \times 3 \times 7.$$

Therefore:

$$\frac{30}{42} = \frac{2 \times 3 \times 5}{2 \times 3 \times 7}.$$

Cancel the common factors 2 and 3:

$$\frac{30}{42} = \frac{5}{7}.$$

Hence:

$$\frac{5}{7}$$

23. Simplifying Improper Fractions

Improper fractions can also be simplified before converting them into mixed numbers.

Example

Simplify:

$$\frac{24}{18}$$

The GCF of 24 and 18 is 6.

Therefore:

$$\frac{24}{18} = \frac{4}{3}$$

Simplify again:

$$\frac{4}{3} = \frac{2}{3}$$

Thus:

$$\frac{24}{18} = \frac{4}{3}$$

Now convert to a mixed number:

$$4 \div 3 = 1 \text{ remainder } 1.$$

Therefore:

$$\frac{24}{18} = \frac{4}{3} = 1\frac{1}{3}$$

24. Fractions Equal to Zero

A fraction with numerator zero has value zero, provided the denominator is not zero.

For example:

$$\frac{0}{7} = 0.$$

Similarly:

$$\frac{0}{100} = 0.$$

In general:

$$\boxed{\frac{0}{a} = 0}$$

where

$$a \neq 0.$$

25. Exam-Focused Worked Examples

This section combines the main examination patterns from this lecture.

Exam Practice 1

Classify:

$$\frac{5}{12}.$$

Since:

$$5 < 12,$$

the numerator is smaller than the denominator.

Therefore:

Proper Fraction

Exam Practice 2

Classify:

$$\frac{19}{7}.$$

Since:

$$19 > 7,$$

the fraction is improper.

Therefore:

Improper Fraction

Exam Practice 3

Convert:

$$\frac{29}{6}$$

into a mixed number.

Divide:

$$29 \div 6.$$

Since:

$$6 \times 4 = 24,$$

the remainder is:

$$29 - 24 = 5.$$

Therefore:

$$\frac{29}{6} = 4\frac{5}{6}$$

Exam Practice 4

Convert:

$$6\frac{2}{5}$$

into an improper fraction.

$$6 \times 5 = 30.$$

Add 2:

$$30 + 2 = 32.$$

Therefore:

$$6\frac{2}{5} = \frac{32}{5}$$

Exam Practice 5

Find a fraction equivalent to

$$\frac{4}{9}$$

having denominator 45.

Since:

$$9 \times 5 = 45,$$

multiply the numerator by 5:

$$4 \times 5 = 20.$$

Therefore:

$$\frac{4}{9} = \frac{20}{45}$$

Exam Practice 6

Simplify:

$$\frac{45}{60}.$$

The GCF is:

$$15.$$

Therefore:

$$\frac{45}{60} = \frac{45 \div 15}{60 \div 15} = \frac{3}{4}.$$

Hence:

$$\frac{3}{4}$$

Exam Practice 7

Determine whether:

$$\frac{8}{12} \quad \text{and} \quad \frac{10}{15}$$

are equivalent.

Cross multiply:

$$8 \times 15 = 120$$

and:

$$12 \times 10 = 120.$$

Therefore:

$$120 = 120.$$

Hence:

$$\frac{8}{12} = \frac{10}{15}$$

Exam Practice 8

Find x :

$$\frac{7}{8} = \frac{56}{x}.$$

Cross multiply:

$$7x = 8 \times 56.$$

Therefore:

$$7x = 448.$$

So:

$$x = 64.$$

Hence:

$$\boxed{x = 64}$$

26. Common Mistakes

Students should carefully avoid the following errors.

Mistake 1: Wrong Classification

A student may say:

$$\frac{7}{9}$$

is improper because both numbers are large.

This is incorrect.

Classification depends on the relationship between numerator and denominator.

Since:

$$7 < 9,$$

we have:

$$\frac{7}{9} \text{ is proper.}$$

Mistake 2: Mixed Number Conversion

Incorrect:

$$3\frac{2}{5} = \frac{3+2}{5} = \frac{5}{5}.$$

Correct:

$$3\frac{2}{5} = \frac{3 \times 5 + 2}{5} = \frac{17}{5}.$$

Always multiply the whole number by the denominator first.

Mistake 3: Changing Only the Numerator

A student may write:

$$\frac{3}{5} = \frac{6}{5}.$$

This is incorrect.

To form an equivalent fraction, multiply both numerator and denominator by the same number:

$$\frac{3}{5} = \frac{3 \times 2}{5 \times 2} = \frac{6}{10}.$$

Mistake 4: Adding the Same Number

Students sometimes think:

$$\frac{2}{3} = \frac{2+1}{3+1} = \frac{3}{4}.$$

This is false.

In fact:

$$\frac{2}{3} \neq \frac{3}{4}.$$

Equivalent fractions are generated by multiplying or dividing the numerator and denominator by the same nonzero number, not by adding the same number.

Mistake 5: Stopping Before Lowest Form

A student may simplify:

$$\frac{24}{36} = \frac{12}{18}$$

and stop.

But:

$$\frac{12}{18}$$

can still be simplified.

The final lowest form is:

$$\frac{24}{36} = \frac{2}{3}.$$

Mistake 6: Zero Denominator

The fraction

$$\frac{5}{0}$$

is not equal to zero.

It is **undefined** because division by zero is not allowed.

However:

$$\frac{0}{5} = 0.$$

27. Quick Revision Rules

Key Rules to Remember

Rule 1: Proper Fraction

$$a < b \Rightarrow \frac{a}{b} \text{ is proper.}$$

Rule 2: Improper Fraction

$$a \geq b \Rightarrow \frac{a}{b} \text{ is improper.}$$

Rule 3: Improper to Mixed

Divide numerator by denominator.

$$a = bq + r$$

then:

$$\frac{a}{b} = q\frac{r}{b}.$$

Rule 4: Mixed to Improper

$$a\frac{b}{c} = \frac{ac + b}{c}$$

Rule 5: Equivalent Fractions

$$\frac{a}{b} = \frac{ak}{bk}$$

for $k \neq 0$.

Rule 6: Checking Equivalence

$$\frac{a}{b} = \frac{c}{d}$$

when:

$$ad = bc.$$

Rule 7: Lowest Form

Divide numerator and denominator by their Greatest Common Factor.

28. Class Activity

Solve the following problems during the class.

Part A: Identify the Type

Classify each as proper, improper, or mixed.

1. $\frac{3}{8}$

2. $\frac{11}{7}$

3. $2\frac{1}{5}$

4. $\frac{9}{9}$

5. $\frac{17}{20}$

6. $4\frac{3}{7}$

Part B: Improper to Mixed

Convert the following into mixed numbers:

1. $\frac{7}{4}$

2. $\frac{11}{3}$

3. $\frac{17}{5}$

4. $\frac{29}{8}$

5. $\frac{41}{6}$

6. $\frac{83}{11}$

Part C: Mixed to Improper

Convert the following into improper fractions:

1. $1\frac{3}{4}$

2. $2\frac{2}{5}$

3. $3\frac{7}{8}$

4. $5\frac{1}{6}$

5. $7\frac{4}{9}$

Part D: Equivalent Fractions

Complete the following:

$$\frac{1}{2} = \frac{\square}{8}$$

$$\frac{3}{5} = \frac{12}{\square}$$

$$\frac{4}{7} = \frac{\square}{35}$$

$$\frac{5}{6} = \frac{25}{\square}$$

$$\frac{8}{9} = \frac{\square}{45}$$

Part E: Simplification

Reduce each fraction to its lowest form.

1. $\frac{8}{12}$

2. $\frac{18}{24}$

3. $\frac{25}{35}$

4. $\frac{36}{48}$

5. $\frac{42}{56}$

6. $\frac{63}{81}$

29. Practice Problems

A. Basic Questions

1. Define a proper fraction.
2. Define an improper fraction.
3. Define a mixed number.
4. What is the numerator of $\frac{13}{17}$?
5. What is the denominator of $\frac{11}{15}$?
6. Why can a denominator not be zero?

B. Conversion Questions

1. Convert $\frac{19}{4}$ to a mixed number.
2. Convert $\frac{35}{8}$ to a mixed number.
3. Convert $4\frac{3}{5}$ to an improper fraction.
4. Convert $6\frac{2}{7}$ to an improper fraction.
5. Convert $10\frac{3}{11}$ to an improper fraction.

C. Equivalent Fraction Questions

Find the missing value.

1. $\frac{2}{3} = \frac{x}{15}$
2. $\frac{4}{7} = \frac{20}{x}$
3. $\frac{5}{9} = \frac{x}{54}$
4. $\frac{7}{10} = \frac{35}{x}$
5. $\frac{11}{12} = \frac{x}{60}$

D. Simplification Questions

Simplify:

1. $\frac{14}{21}$

2. $\frac{24}{32}$

3. $\frac{45}{75}$

4. $\frac{56}{72}$

5. $\frac{84}{108}$

6. $\frac{120}{150}$

E. Higher-Level Questions

1. Are

$$\frac{12}{18} \quad \text{and} \quad \frac{20}{30}$$

equivalent? Justify your answer.

2. A student says:

$$\frac{3}{5} = \frac{4}{7}$$

because both numerator and denominator were increased by 1. Explain why the statement is incorrect.

3. Find the simplest fraction equivalent to

$$\frac{72}{96}.$$

4. Convert

$$5\frac{7}{12}$$

into an improper fraction.

5. Convert

$$\frac{101}{12}$$

into a mixed number.

30. Home Assignment

1. Classify the Fractions

State whether each is proper, improper or mixed.

1. $\frac{7}{12}$

2. $\frac{15}{11}$

3. $3\frac{4}{9}$

4. $\frac{13}{13}$

5. $\frac{21}{25}$

2. Improper to Mixed Number

Convert:

1. $\frac{15}{4}$

2. $\frac{23}{6}$

3. $\frac{37}{8}$

4. $\frac{52}{9}$

5. $\frac{79}{12}$

3. Mixed Number to Improper Fraction

Convert:

1. $2\frac{1}{3}$

2. $4\frac{5}{7}$

3. $6\frac{3}{8}$

4. $8\frac{5}{11}$

5. $10\frac{7}{12}$

4. Equivalent Fractions

Find the missing value.

1.

$$\frac{3}{4} = \frac{x}{20}$$

2.

$$\frac{5}{8} = \frac{30}{x}$$

3.

$$\frac{7}{12} = \frac{x}{60}$$

4.

$$\frac{9}{11} = \frac{45}{x}$$

5. Simplification

Reduce each fraction to its lowest form.

1. $\frac{20}{30}$

2. $\frac{36}{54}$

3. $\frac{48}{64}$

4. $\frac{75}{100}$

5. $\frac{96}{144}$

End of Lecture 02

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